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from sklearn.preprocessing import MinMaxScaler, StandardScaler,
RobustScaler, QuantileTransformer, PowerTransformer
import numpy as np
fare = df['Fare']
# 2. Min-Max
scaler_minmax = MinMaxScaler()
fare_minmax = scaler_minmax.fit_transform(fare.values.reshape(-1, 1))

# 3. Z-Score Normalization
scaler_zscore = StandardScaler()
fare_zscore = scaler_zscore.fit_transform(fare.values.reshape(-1, 1))

# 4. Decimal Scaling
j = len(str(int(fare.abs().max())))
fare_decimal = fare / (10**j)

# Combine results into DataFrame
df_transformed = pd.DataFrame({
    'Original Fare': fare,
    'Min-Max': fare_minmax.flatten(),
    'Z-Score': fare_zscore.flatten(),
    'Decimal Scaling': fare / (10**j),
})
print(df_transformed)

# Plot semua hasil transformasi
fig, axes = plt.subplots(2, 2, figsize=(16, 12))
axes = axes.flatten()

plot_titles = df_transformed.columns.tolist()
for i in range(len(plot_titles)):
    sns.histplot(df_transformed[plot_titles[i]], kde=True, ax=axes[i])
    axes[i].set_title(plot_titles[i])

# hapus bagian plot yang tidak ada isinya
if len(plot_titles) < len(axes):
    for j in range(len(plot_titles), len(axes)):
        fig.delaxes(axes[j])

plt.tight_layout()
plt.show()

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import seaborn as sns
import matplotlib.pyplot as plt

# Visualisasi Age
plt.figure(figsize=(10, 4))
sns.histplot(df['Age'], kde=True, color='skyblue')

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plt.title('Distribusi Usia')
plt.show()

# Visualisasi Fare
plt.figure(figsize=(10, 4))
sns.histplot(df['Fare'], kde=True, color='salmon')
plt.title('Distribusi Tarif (Fare)')
plt.show()

# Cek Skewness secara cepat
print(df[['Age', 'Fare']].skew())
```

```
from sklearn.preprocessing import QuantileTransformer

# Pastikan fare adalah pandas Series
fare = pd.Series(fare)

# 1. Log Transformation
fare_log = np.log1p(fare)

# 2. Quantile Transformer (Normal distribution)
qt = QuantileTransformer(output_distribution='normal', random_state=42)
fare_qt = qt.fit_transform(fare.values.reshape(-1, 1))

# Ubah hasil QT ke pandas Series agar bisa pakai .skew()
fare_qt_series = pd.Series(fare_qt.flatten())

# 3. Gabungkan ke DataFrame
df_transformed2 = pd.DataFrame({
    'Log Transform': fare_log,
    'Quantile Normal': fare_qt_series
})

# 4. Plot
fig, axes = plt.subplots(1, 2, figsize=(16, 6))

for i, col in enumerate(df_transformed2.columns):
    sns.histplot(df_transformed2[col], kde=True, ax=axes[i])
    axes[i].set_title(col)

plt.tight_layout()
plt.show()

# 5. Hitung skewness
print("Skew Log Transform:", fare_log.skew())
print("Skew Quantile Normal:", fare_qt_series.skew())
```

```
# #6. hitung kurtosis
# print("Kurtosis Log Transform:", fare_log.kurt())
# print("Kurtosis Quantile Normal:", fare_qt_series.kurt())
```

```
from sklearn.preprocessing import LabelEncoder
#LABEL ENCODE
le = LabelEncoder()
df['Sex_encoded'] = le.fit_transform(df['Sex'])
df.drop(['Sex'], axis=1, inplace=True)
```

```
#ONE HOT ENCODE
df = pd.get_dummies(df, columns=['Embarked'], drop_first=True, dtype=float)
```